P1 Particles and Radiation – Knowledge Organiser: Self-Test

AQA A-level Physics • Set 1: Constituents of the Atom **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Charge of a proton				$+1.60 \times 10^{-19}$ C
2	Charge of an electron				-1.60×10^{-19} C
3	Mass of a proton				1.67×10^{-27} kg
4	Mass of an electron				9.11×10^{-31} kg
5	Specific charge				Charge ÷ mass (unit C kg^{-1})
6	Specific charge of a proton				9.58×10^{7} C kg^{-1}
7	Specific charge of an electron				1.76×10^{11} C kg^{-1}
8	Proton number (*Z*)				Number of protons
9	Nucleon number (*A*)				Number of protons + neutrons
10	Isotopes				Same number of protons, different number of neutrons

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

P1 Particles and Radiation – Knowledge Organiser: Self-Test

AQA A-level Physics • Set 2: Stable and Unstable Nuclei **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Role of the strong nuclear force				Holds the nucleus together, overcoming electrostatic repulsion between protons
2	Range over which the strong force is attractive				Up to about 3 fm
3	Strong force at separations below about 0.5 fm				Repulsive
4	Nuclei that undergo α decay				Very large nuclei (too many nucleons)
5	Nuclei that undergo β^- decay				Neutron-rich nuclei
6	What an α particle is				2 protons + 2 neutrons (a helium nucleus)
7	Change in *A* and *Z* in α decay				*A* decreases by 4, *Z* decreases by 2
8	Change in *A* and *Z* in β^- decay				*A* unchanged, *Z* increases by 1
9	Evidence that led to the neutrino being proposed				β particles are emitted with a range of energies
10	Why the neutrino was needed				To conserve energy (and momentum) in β decay

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

P1 Particles and Radiation – Knowledge Organiser: Self-Test

AQA A-level Physics • Set 3: Particles, Antiparticles and Photons **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Antiparticle of the electron				Positron
2	How an antiparticle compares with its particle				Same mass and rest energy, opposite charge
3	Rest energy of an electron				0.511 MeV
4	Rest energy of a proton				938 MeV
5	Photon energy equation				$E = hf = hc/\lambda$
6	Converting MeV to J				$\times 1.60 \times 10^{-13}$
7	Annihilation				Particle + antiparticle → two photons
8	Why annihilation produces two photons				To conserve momentum
9	Pair production				Photon → particle + antiparticle
10	Minimum photon energy for pair production				Total rest energy of the particle and antiparticle

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

P1 Particles and Radiation – Knowledge Organiser: Self-Test

AQA A-level Physics • Set 4: Particle Interactions **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Exchange particle for the electromagnetic force				Virtual photon
2	Exchange particles for the weak interaction				W^+ and W^- bosons
3	Exchange particle for the strong force between nucleons				Pion
4	Particles the strong interaction acts on				Hadrons only
5	Interaction responsible for β decay				Weak interaction
6	Exchange particle in β^- decay				W^-
7	Exchange particle in β^+ decay				W^+
8	Equation for β^- decay				$n \rightarrow p + e^- + \bar{\nu}_e$
9	Equation for β^+ decay				$p \rightarrow n + e^+ + \nu_e$
10	Equation for electron capture				$p + e^- \rightarrow n + \nu_e$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

P1 Particles and Radiation – Knowledge Organiser: Self-Test

AQA A-level Physics • Set 5: Classification of Particles **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Hadrons				Particles that feel the strong interaction
2	Two classes of hadron				Baryons and mesons
3	Quark structure of a baryon				Three quarks
4	Quark structure of a meson				Quark + antiquark
5	The only stable baryon				Proton
6	Baryon number of a baryon, antibaryon and meson				+1, −1, 0
7	Products of free neutron decay				Proton, electron and antineutrino
8	Two types of meson in the AQA spec				Pions and kaons
9	Leptons				Particles that do not feel the strong interaction (e.g. electron, muon, neutrinos)
10	What a muon decays into				Electron (+ neutrinos)

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

P1 Particles and Radiation – Knowledge Organiser: Self-Test

AQA A-level Physics • Set 6: Quarks and Antiquarks **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Charge of an up quark				$+\frac{2}{3}e$
2	Charge of a down quark				$-\frac{1}{3}e$
3	Charge of a strange quark				$-\frac{1}{3}e$
4	Baryon number of a quark				$+\frac{1}{3}$
5	Strangeness of a strange quark				-1
6	Quark structure of a proton				uud
7	Quark structure of a neutron				udd
8	Quark structure of a π^+				$u\bar{d}$
9	Quark structure of a K^+				$u\bar{s}$
10	Quark change in β^- decay				$d \rightarrow u$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

P1 Particles and Radiation – Knowledge Organiser: Self-Test

AQA A-level Physics • Set 7: Conservation Laws **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Quantities conserved in every particle interaction				Charge, baryon number, lepton number (plus energy and momentum)
2	Electron lepton number of an electron				$L_e = +1$
3	Electron lepton number of a positron				$L_e = -1$
4	Muon lepton number of a muon neutrino				$L_\mu = +1$
5	Baryon number of a lepton				0
6	Quantity conserved in strong but not weak interactions				Strangeness
7	Allowed change in strangeness in a weak interaction				0, +1 or −1
8	Why strange particles are produced in pairs				Made by the strong interaction, which conserves strangeness
9	Interaction by which strange particles decay				Weak
10	What kaons decay into				Pions (or leptons)

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

P1 Particles and Radiation – Knowledge Organiser: Self-Test

AQA A-level Physics • Set 8: The Photoelectric Effect **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Photoelectric effect				Electrons emitted from a metal surface when light above a threshold frequency shines on it
2	Threshold frequency				Minimum frequency of light that causes electrons to be emitted
3	Work function (φ)				Minimum energy needed to remove an electron from the metal surface
4	Effect of increasing intensity above the threshold frequency				More electrons emitted per second; $E_{k(max)}$ unchanged
5	Effect of increasing intensity below the threshold frequency				Still no electrons emitted
6	Effect of increasing frequency				$E_{k(max)}$ increases
7	Time delay before emission				None (emission is instant)
8	Why the wave theory cannot explain the effect				It predicts emission at any frequency if the light is intense enough
9	How the photon model explains the threshold frequency				One photon gives all its energy to one electron, so hf must be ≥ φ
10	Scientist who explained the effect using photons				Einstein

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

P1 Particles and Radiation – Knowledge Organiser: Self-Test

AQA A-level Physics • Set 9: Photoelectric Equation and Stopping Potential **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Photoelectric equation				$hf = \varphi + E_{k(max)}$
2	Threshold frequency equation				$f_0 = \varphi/h$
3	Why the kinetic energy is a maximum				Electrons below the surface lose energy escaping
4	Stopping potential (V_s)				Minimum p.d. needed to stop the fastest photoelectrons
5	Equation linking stopping potential and $E_{k(max)}$				$eV_s = E_{k(max)}$
6	Electronvolt (eV)				Energy gained by an electron accelerated through a p.d. of 1 V
7	1 eV in joules				1.60×10^{-19} J
8	Gradient of an $E_{k(max)}$ against f graph				h
9	x-intercept of an $E_{k(max)}$ against f graph				Threshold frequency f_0
10	y-intercept of an $E_{k(max)}$ against f graph				$-\varphi$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

P1 Particles and Radiation – Knowledge Organiser: Self-Test

AQA A-level Physics • Set 10: Energy Levels and Spectra **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Excitation				An electron moves up to a higher energy level
2	Ionisation				An electron is removed from the atom
3	Ground state				Lowest energy level of an atom
4	Energy a photon needs to excite an atom				Exactly equal to the difference between two energy levels
5	Energy a colliding electron needs to excite an atom				Kinetic energy at least equal to the energy difference
6	Photon emitted when an electron moves down a level				$hf = E_1 - E_2$
7	How mercury atoms are excited in a fluorescent tube				Collisions with free electrons accelerated by the p.d.
8	Job of the phosphor coating				Absorbs UV photons and emits visible light
9	Line absorption spectrum				Dark lines on a continuous spectrum
10	What line spectra show				Electrons in atoms have discrete energy levels

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

P1 Particles and Radiation – Knowledge Organiser: Self-Test

AQA A-level Physics • Set 11: Wave–Particle Duality **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Evidence that light behaves as a wave				Diffraction and interference
2	Evidence that light behaves as a particle				Photoelectric effect
3	Evidence that electrons behave as waves				Electron diffraction
4	de Broglie equation				$\lambda = h/mv$
5	Quantity mv				Momentum
6	Effect of increasing electron speed on the diffraction pattern				Rings get closer together (λ decreases)
7	How electron speed is changed in an electron diffraction tube				Change the accelerating p.d.
8	Condition for significant diffraction				Wavelength similar to the gap or atomic spacing
9	Electron or proton at the same speed: longer λ				Electron (smaller mass)
10	Why a thrown ball shows no diffraction				Its wavelength is far too small

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.